ANALYSING THE EFFECT OF NETWORK DEPTH ON MULTILAYER PERCEPTRON (MLP) PERFORMANCE

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GitHub: https://github.com/Bhavana-Prasad/mlp-depth-analysis.git

1. Introduction:

Machine Learning (ML) is the part of artificial intelligence where the machine learners from the data and makes predictions or decisions without being explicitly programmed. Among all the machine learning models, neural networks are gaining significance due to their ability to handle complex relationships.

Multilayer Perceptron (MLP) is the feedforward neural network with one or more hidden layers and an output layer. The important factor to consider for MLP is depth, which is the number of hidden layers present in the MLP.

In the given tutorial, we will discuss the effect of the depth of the MLP on its learning ability, and also how the increase in the depth of the MLP will result in overfitting or not.

2. Background and Theory:

An MLP is a series of interconnected neurons, each of which performs a weighted sum of inputs, adds a bias, and then uses an activation function. The network learns by adjusting the weights using optimization algorithms like gradient descent.

Hidden Layers and Depth:

The layers that are "hidden" in an MLP allow for learning representations of data. Increasing the number of "hidden layers" increases the capacity of the MLP to learn complex patterns.

- A shallow MLP (few layers) might be unable to capture complex patterns.
- A deep MLP (many layers) can capture complex patterns but can also learn noise.

Activation Function:

Activation functions provide non-linearity. In this study, ReLU (Rectified Linear Unit) is used as an activation function. ReLU returns 0 if the input is negative and returns a linear function of the input if it is positive.

Underfitting and Overfitting:

- Underfitting is when a model is too simple to learn the underlying structure of the data.
- Overfitting is when a model is too complex and learns all the noise in the training data, hence performing badly on new data.

The aim is to achieve an optimal balance.

3. Methodology:

To examine the effect of network depth, three different MLP models of varying depths were trained and tested. A fixed random state was used to guarantee reproducibility of results.

Dataset:

The dataset was created using a synthetic dataset provided by using the make_moons function. It is a commonly used dataset to test classification algorithms since it is not linearly separable.

Number of samples: 300

Noise level: 0.2

Data Preparation:

The dataset was divided into a training set and a testing set:

Training set: 70%

Testing set: 30%

A constant random_state is used to make results comparable across multiple runs.

Model Configuration:

Three MLP models with varying depths were defined to check the impact of depth on model performance:

Model	No. of Hidden Layers
Shallow	(10,)
Medium	(10, 10)
Deep	(10,10,10,10,10)

All models are defined with:

Activation Function = ReLU

Maximum iterations = 2000

Evaluation Metrics:

The performance of all these models is measured using:

Training Accuracy: It measures how well a model can fit the training data

Testing Accuracy: It measures how well a model can generalize and how well it is able to perform on new data.

4. Experimental Results:

Dataset Visualization:

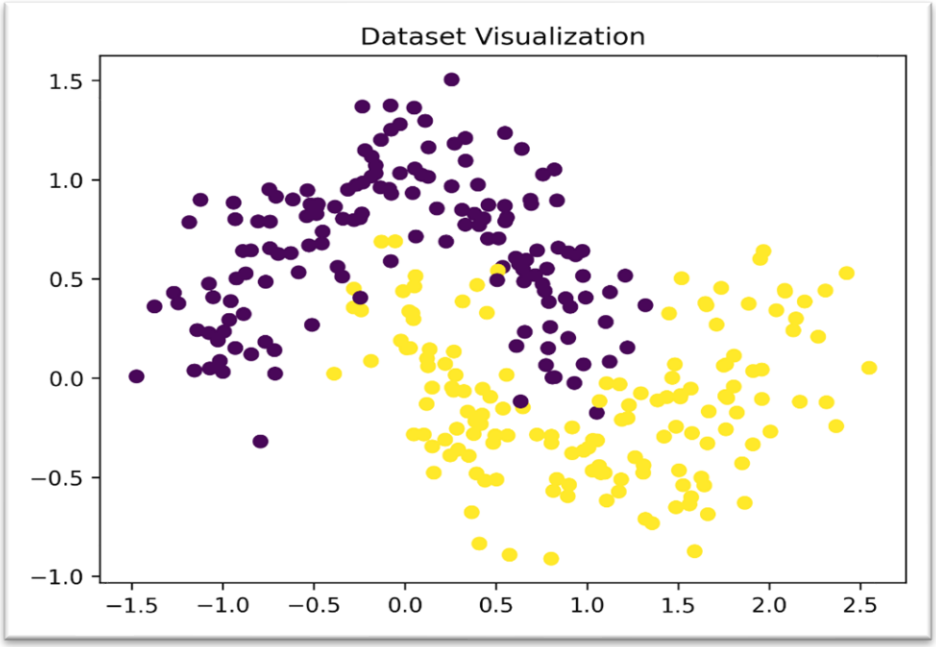


Figure 1: Visualization of the synthetic data created using the make_moons function. This data follows a non-linear pattern that needs to be correctly classified by a flexible model.

Accuracy Comparison:

Model	Train Accuracy	Test Accuracy
Shallow	0.871	0.889
Medium	0.895	0.911
Deep	0.967	0.967

The results show that there is a significant improvement in performance when the depth of the network is increased.

Decision Boundary Analysis:

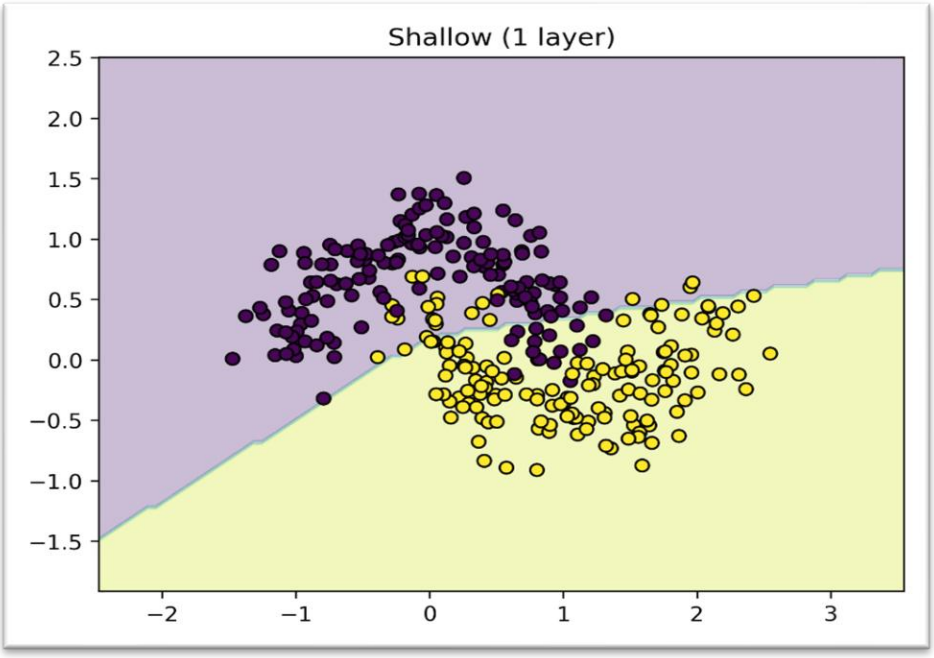


Figure 2: Decision boundary for shallow MLP (1 hidden layer)

From the decision boundary, it is clear that the shallow model has performed poorly in producing an accurate decision boundary. The decision boundary is simple and does not take into account the complexity of the data.

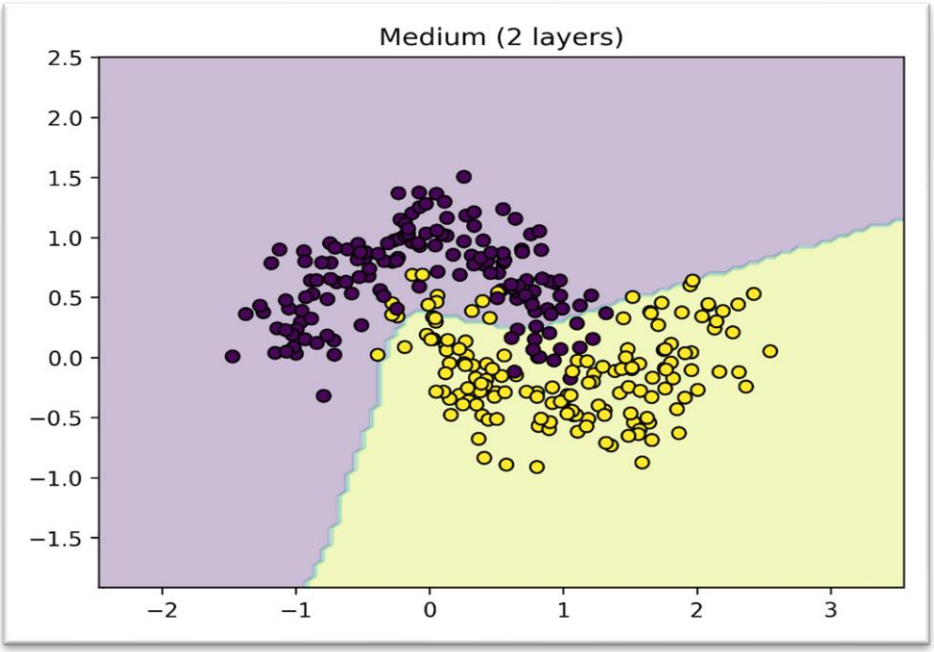


Figure 3: Decision boundary for medium MLP (2 hidden layers)

From the decision boundary, it is clear that the performance of the model is good. The model has managed to learn the complexity of the data and produce an accurate decision boundary.

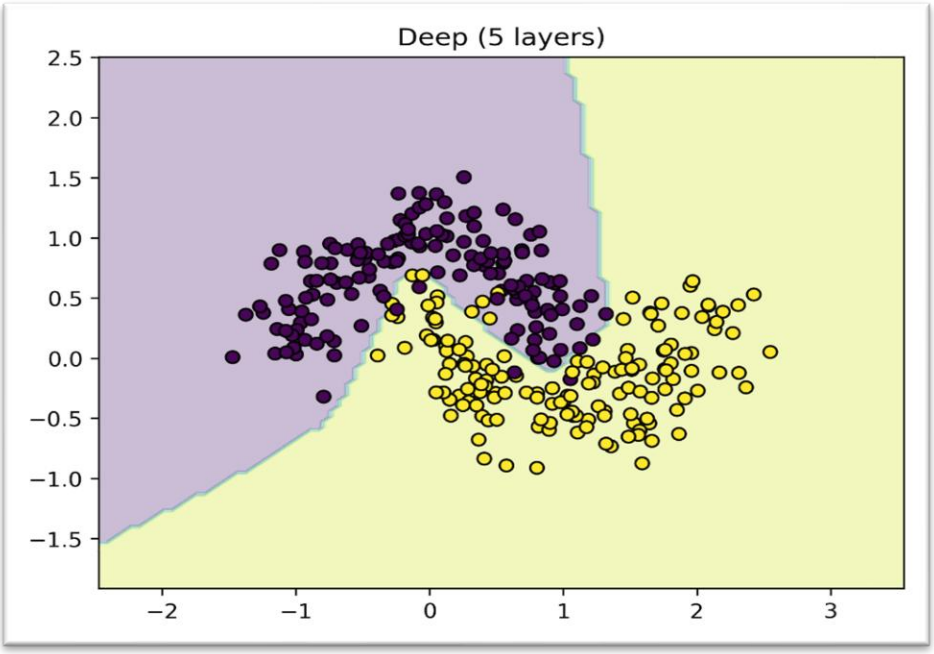
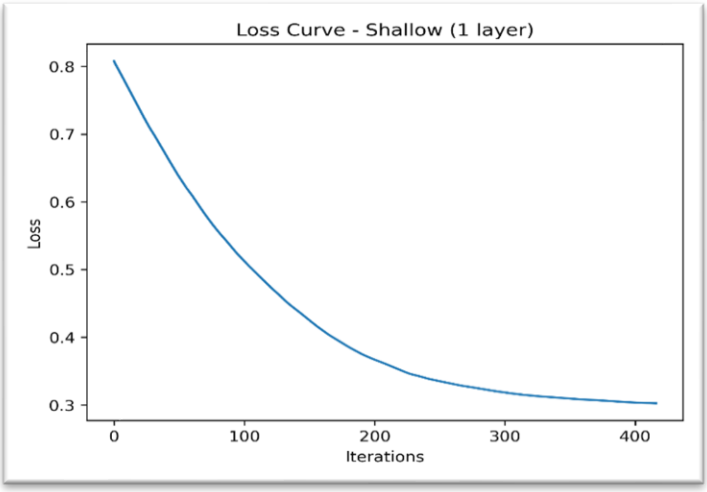
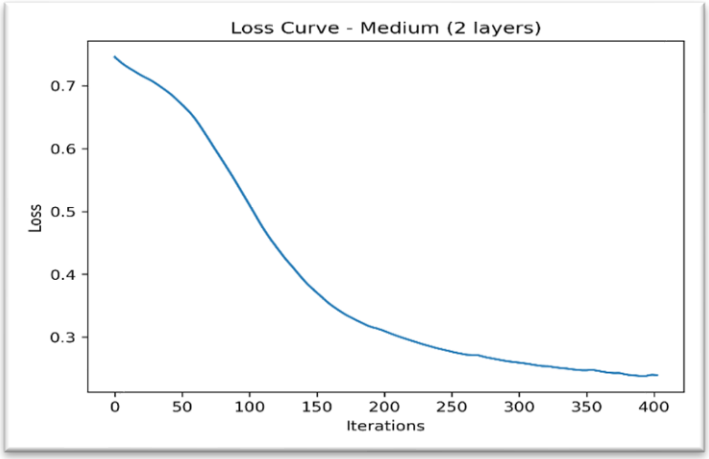


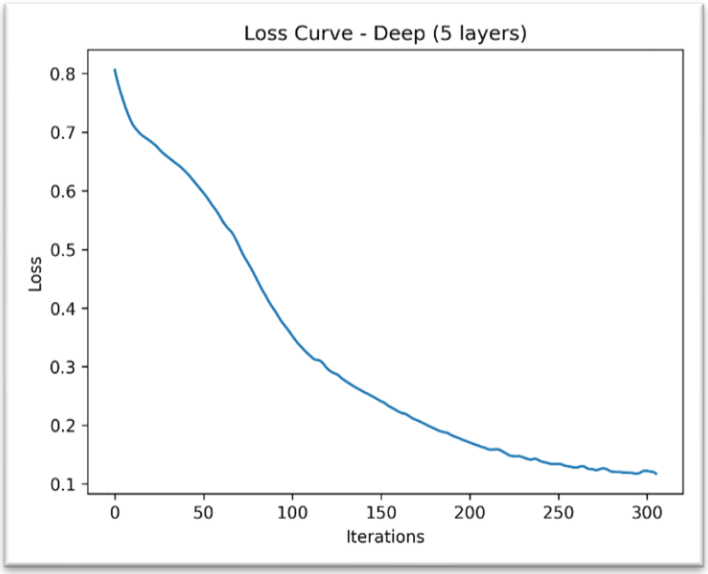
Figure 4: Decision boundary for deep MLP (5 hidden layers)

From the decision boundary, the model has performed very well in producing an accurate decision boundary. The decision boundary is very refined.

Loss Curve Analysis:







Figures: Loss curves for models with different depths.

The loss curves indicate that the models are converging during the training process. However, the loss of the deeper model is lower, showing that it was optimized and learned well.

5. Discussion:

The experimental results demonstrate that depth is a critical factor that affects model performance.

The performance of the shallow model indicates that it is underfitting because it cannot learn the non-linear relationships that are present in the dataset. This is also evidenced by its poor accuracy on both the training and testing sets.

The performance of the medium model is better. This shows that depth is an important factor that affects model performance. This model can learn complex relationships that are present in the dataset.

The performance of the deep model is better than that of both the shallow and medium models. This shows that this model can learn complex relationships that are present in the dataset. It is also interesting to note that the accuracy of this model on both the training and testing sets is approximately equal. This shows that this model is not overfitting.

The reason for this is that this dataset is relatively simple. Although deep models are more prone to overfitting, this dataset does not have any noise.

6. Conclusion:

This paper sought to investigate the impact of the depth of the network on the performance of a Multilayer Perceptron. From the analysis, it was established that the depth of the network has a positive effect on the ability of the network to learn.

However, it was also established that the increase in the depth of the network may not always result in the occurrence of overfitting, as was seen in the experiment. This was seen to be dependent on the nature of the dataset used.

In the real world, it is crucial to conduct the necessary experiments to achieve the desired results by selecting the most suitable depth for the network.

7. References:

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- Bishop, C.M. (2006) *Pattern Recognition and Machine Learning*. New York: Springer.
- Scikit-learn (n.d.) Scikit-learn: Machine Learning in Python. Available at: https://scikit-learn.org (Accessed: 25 March 2026).
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